

Computer Networks

Assignment # 02

Subnetting

Name : Abdullah Ayub

Roll NO : BSDSF24MD29

Question # 1 : VLSM for 192.168.50.0/24

we would sort departments first from largest to smallest

Departments	Hosts Needed	Subnet	CIDR	Useable Hosts
Admin	60	/26	64	62
HR	30	/27	32	30
Sales	28	/27	32	30
IT	12	/28	16	14
Security	6	/29	8	6

Allocate From 192.168.50.0 :

Admin (60 Hosts)

Nw : 192.168.50.0/26
 Subnet Mask : 255.255.255.192
 Broadcast : 192.168.50.63
 First Host : 192.168.50.1
 Last : 192.168.50.62

HR (30 Hosts)

192.168.50.64/27
 255.255.255.224
 192.168.50.95
 192.168.50.65
 192.168.50.94

Sales (28 Hosts)

192.168.50.96/27
 255.255.255.224
 192.168.50.127
 192.168.50.97
 192.168.50.126

IT (12 Hosts)

Nw : 192.168.50.128/28
 Subnet : 255.255.255.240
 Broadcast : 192.168.50.143
 First : 192.168.50.129
 Last : 192.168.50.142

Security (6 Hosts)

192.168.50.144/29
 255.255.255.248
 192.168.50.151
 192.168.50.145
 192.168.50.150

Question #2: CIDR Subnetting

Given: Network: 10.10.0.0/22
Required: 4 equal subnets

New subnet size :

we need 4 subnets so,

$$4 = 2^2 \text{ borrow 2 bits}$$

• Original: /22

• new: /24

Each subnet :

• Total IPS : 256

• Useable IPS : 254

Head office

Subnet No: 1
NW Address: 10.10.0.0
CIDR: /24
Broadcast: 10.10.0.255
First IP: 10.10.0.1
Last IP: 10.10.0.254

Lahore Branch

2
10.10.1.0
/24
10.10.1.255
10.10.1.1
10.10.1.254

Karachi Branch

3
10.10.2.0
/24
10.10.2.255
10.10.2.1
10.10.2.254

Islamabad Branch

SubNet NO 4
NW Addr: 10.10.3.0
CIDR /24
Broadcast 10.10.3.255
First 10.10.3.1
last: 10.10.3.254